

A PILOT STUDY DETAILS

To ground our research in real user experiences, we conducted a formative pilot study ($N = 12$) to examine how AR novices interact with existing anchoring techniques during movement and multitasking in a mobile remote collaboration scenario.

During a 40-minute session, participants completed a “walking–reading–communicating” task to compare head- (i.e., relative to head orientation), body- (i.e., relative to chest), and world-locked (i.e., relative to a fixed environment location) panels. For each technique, they walked down a 30-foot cluttered hallway while reading from a floating virtual panel. At two checkpoints, they stopped to read physical questions posted on the wall and verbally answered them via an audio call with the experimenter. This scenario required continuous attention shifting between physical navigation, virtual content, physical information processing, and communication. Following the task, participants took part in semi-structured interviews.

We transcribed semi-structured post-task interviews into observation notes and analyzed them using affinity diagramming¹. **Tab. 1** summarizes the key findings.

Table 1: Perceived trade-offs of conventional anchoring techniques (Pilot Study, N=12).

Theme	World-Locked	Body-Locked	Head-Locked
Spatial Predictability	High: Spatially stable, but requires physical return; content shrinks or is left behind when moving away (6/12).	Moderate: Moves with the torso, but sudden direction changes cause perceived drift (5/12).	Low: Moves with every head motion, making its position <i>relative to</i> the environment unpredictable (4/12).
Temporal Availability	Low: Frequently “left behind” during locomotion, requiring users to re-trace their steps to retrieve information (10/12).	Moderate: Follows the user, but chest-level placement obstructs the forward view during stationary physical tasks (4/12).	High: Continuously visible, but persistent presence creates blind spots (in cluttered hallway), safety risks, and cognitive fatigue (10/12).
Attention Allocation	Navigation competes with remembering the panel’s location, reducing cognitive capacity for conversation (5/12).	Stationary obstruction forces users to reposition their bodies to view the physical task, splitting attention (5/12).	Content persistently occupies users forward view (e.g., central vision); users felt “trapped behind a screen” and struggled to disengage (10/12).
Communication & Collaboration	Requires returning to the anchored panel to share information, limiting mobile collaboration (9/12).	Obstructs the forward view at checkpoints, forcing awkward head rotations to read physical materials (4/12).	Central vision occlusion disrupts physical reading and natural conversational cues, forcing reliance on peripheral vision (8/12).

B SUBJECTIVE MEASURES

Tab. 2 shows the measurement items for *Temporal Availability*, and *Spatial Predictability*, following previous works [11, 37, 9, 18].

Table 2: 7-point Likert scale items for subjective experience on *Temporal Availability* (in *Study 1*) and *Spatial Predictability* (in *Study 2*).

Temporal Availability

- The AR content remained visually stable throughout the task and did not drift unexpectedly.
- The visual presentation of the content was steady enough and did not distract me from the task.
- The system’s responses were smooth and natural, without abrupt or confusing movements.

Spatial Predictability

- I could predict where the content would appear after I turned my head or body.
- I was always aware of the direction of the content and never lost track of its location.
- I felt a good sense of control over the position of the AR content.
- After switching my attention to another task, I could quickly relocate the main content.

C PARTICIPANT RESULTS

Tab. 3, **Tab. 4**, and **Tab. 5** present the participant results for *Study 1* and *Study 2*.

Table 3: Summary of *Study 1* significant effects and anchoring values. *Note.* Values are means $\pm$ standard deviations.

Measure	p	<i>AdaptiveLock</i>	<i>ScaleStable</i>	<i>PeriphGuide</i>	<i>AdaptiveFlow</i>
<i>Interface Attention Ratio</i>	< .001	55.85 $\pm$ 12.43	38.63 $\pm$ 13.10	57.79 $\pm$ 14.03	54.52 $\pm$ 13.14
<i>Immediate Recall</i> (higher better)	< .001	10.56 $\pm$ 0.52	8.69 $\pm$ 1.03	10.25 $\pm$ 0.84	10.94 $\pm$ 0.86
<i>Temporal Availability</i>	.013	5.56 $\pm$ 0.41	5.08 $\pm$ 0.50	5.71 $\pm$ 0.37	5.50 $\pm$ 0.38
<i>RTLX</i> (lower better)	< .001	46.22 $\pm$ 6.23	50.67 $\pm$ 4.85	50.55 $\pm$ 8.53	49.67 $\pm$ 6.03
<i>SUS</i> (higher better)	.024	83.75 $\pm$ 12.99	80.83 $\pm$ 15.64	70.42 $\pm$ 10.54	66.67 $\pm$ 12.49

¹Karen Holtzblatt, Jessamyn Burns Wendell, and Shelley Wood. 2005. *Rapid Contextual Design: A How-To Guide to Key Techniques for User-Centered Design*.

Table 4: Summary of *Study 1* significant effects and environment values. *Note.* Values are means $\pm$ standard deviations.

Measure	<i>p</i>	<i>Open Space</i>	<i>Narrow Corridor</i>	<i>Room-to-Room</i>
<i>Interface Attention Ratio</i>	.002	55.41 $\pm$ 11.97	54.25 $\pm$ 16.35	45.43 $\pm$ 15.00
<i>Immediate Recall</i> (higher better)	> .05	10.40 $\pm$ 0.60	9.81 $\pm$ 1.06	10.12 $\pm$ 0.55
<i>Temporal Availability</i>	< .001	5.83 $\pm$ 0.36	4.90 $\pm$ 0.50	5.66 $\pm$ 0.41
<i>RTLX</i> (lower better)	< .001	36.74 $\pm$ 8.11	48.22 $\pm$ 5.70	62.88 $\pm$ 7.10

Table 5: Summary of *Study 2* significant effects and anchoring values. *Note.* Values are means $\pm$ standard deviations.

Measure	<i>p</i>	<i>AdaptiveLock</i>	<i>ScaleStable</i>	<i>PeriphGuide</i>	<i>AdaptiveFlow</i>
<i>Interface Attention Ratio</i>	.032	31.80 $\pm$ 9.42	25.32 $\pm$ 16.03	40.22 $\pm$ 10.78	40.63 $\pm$ 16.55
<i>Spatial Predictability</i>	< .010	4.94 $\pm$ 1.13	5.90 $\pm$ 1.30	4.89 $\pm$ 1.35	4.39 $\pm$ 1.53
<i>RTLX</i> (lower better)	.035	38.60 $\pm$ 16.35	49.21 $\pm$ 19.08	55.41 $\pm$ 14.67	58.62 $\pm$ 12.31
<i>Perceived Collaboration</i> (higher better)	.018	6.40 $\pm$.48	6.52 $\pm$.26	6.32 $\pm$.31	6.30 $\pm$.36